



Acute Febrile illness in Pediatrics

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Outlines

- Overview of malaria
 - Etiology and Epidemiology
 - Lifecycle and pathogenesis
 - Clinical manifestation
 - Diagnosis
 - Management
- Typhoid Fever
- Relapsing Fever
- Endemic Typhus
- Schistosomiasis
- Leishmaniasis

OVERVIEW OF MALARIA

ENVIRONMENT



MALARIA

Definition

- Malaria is a life-threatening disease caused by Plasmodium parasites, which are spread to people through the bites of infected female Anopheles mosquitoes.
- An **AFI** caused by Characterized by paroxysms of fever, chills, sweats, fatigue, anemia, and splenomegaly.
- It is preventable and curable.
- Name is derived from the Italian word
Mal' aria or bad air

Etiology

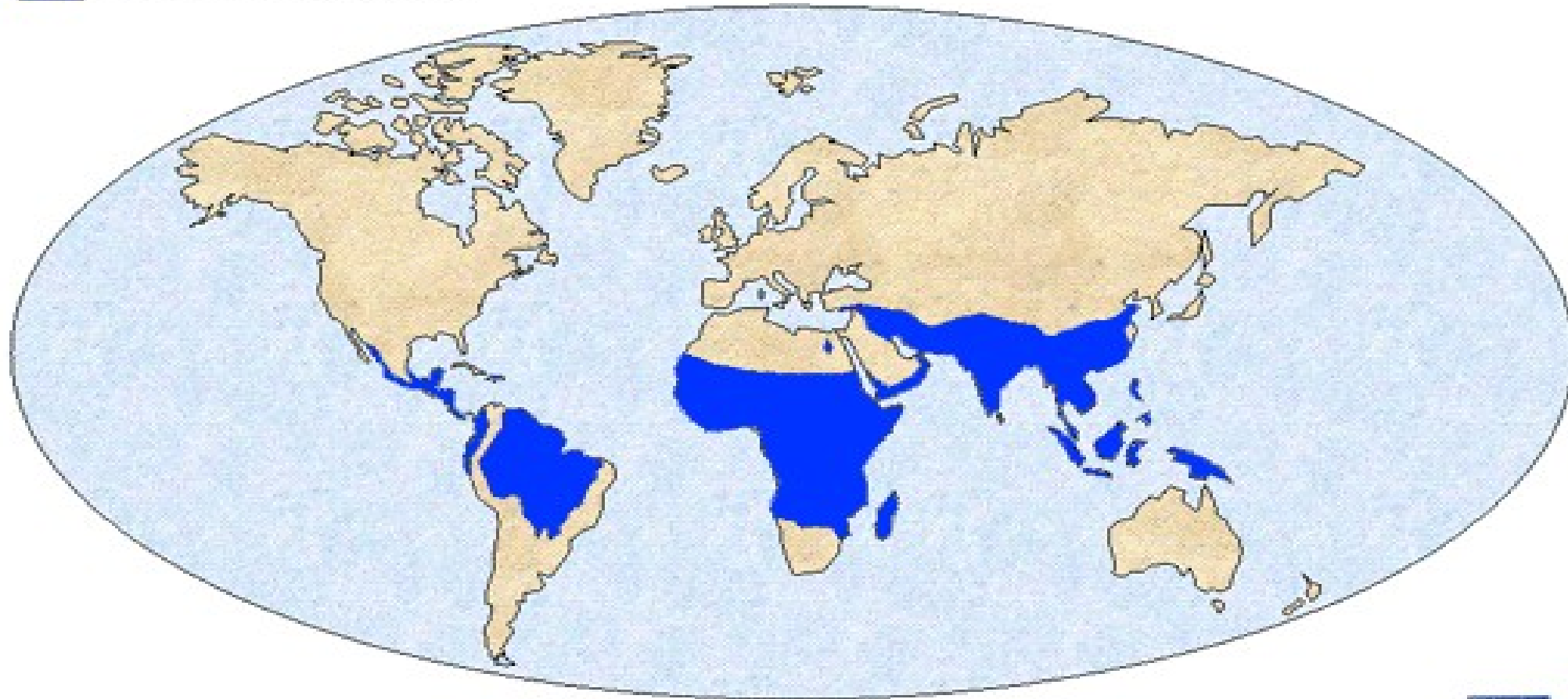
- Malaria is caused by intracellular *Plasmodium* protozoa transmitted to humans by female *Anopheles mosquitoes*.
- **Five species** of *Plasmodium* cause malaria in humans:
 - ✓ *P. falciparum*
 - ✓ *P. malariae*
 - ✓ *P. ovale*
 - ✓ *P. vivax*
 - ✓ *P. knowlesi*(since 2004)
- In Ethiopia, **60%-70%** of malaria cases have been due to *P. falciparum*, with the remainder caused by *P. vivax*.

Epidemiology

- Malaria is endemic throughout the tropical areas of the world
- **In 2020**, a total of **241 million** malaria cases and **627,000** malaria deaths reported globally and
- **About 96%** of deaths occur in Sub-Saharan Africa and **77% of deaths** are children under five (**WHO 2021**).
- The incidence rate of malaria is estimated to have decreased by **37%** globally between 2000 and 2015.
- Malaria death rates have decreased by **60%** over the same period (WHO, 2015).
- Malaria case incidence (i.e. cases per 1000 population at risk) reduced **from 81** in 2000 to 59 in 2015 and **56** in 2019, before increasing again to **59** in **2020**.
- The increase in 2020 was associated with disruption to services during the COVID-19 pandemic (**WHO 2021**)
- Almost all of the deaths are attributable to ***Plasmodium falciparum***.

EPIDEMIOLOGY– WHO, WHEN, WHERE & HOW?

■ Distribution of Malaria



CDC

Epidemiology of Malaria in Ethiopia

- **Malaria** is a leading health problem in Ethiopia.
- *Three quarters (75%)* of the country's landmass is malaria endemic and about **52%** of the total population is at risk of malaria infection (**CSA 2020**).
- **Malaria transmission** in Ethiopia mainly occurs up to the **2000** meter (m) elevation but can also occasionally affect areas up to 2300m elevation.

Cont...

- The *epidemiology of malarial* infections is intricately linked to the distribution and habits of the anopheline vectors in any particular region.
- *In highly endemic areas*, mosquito breeding can take place nearly year-round, and reproductive capacity in the mosquito is maximized by a **tropical climate**.
- In areas where malaria is more of a seasonal phenomenon, its prevalence is related to **rainfall** or other ecologic events that affect the mosquito population.

Trend of malaria in Ethiopia

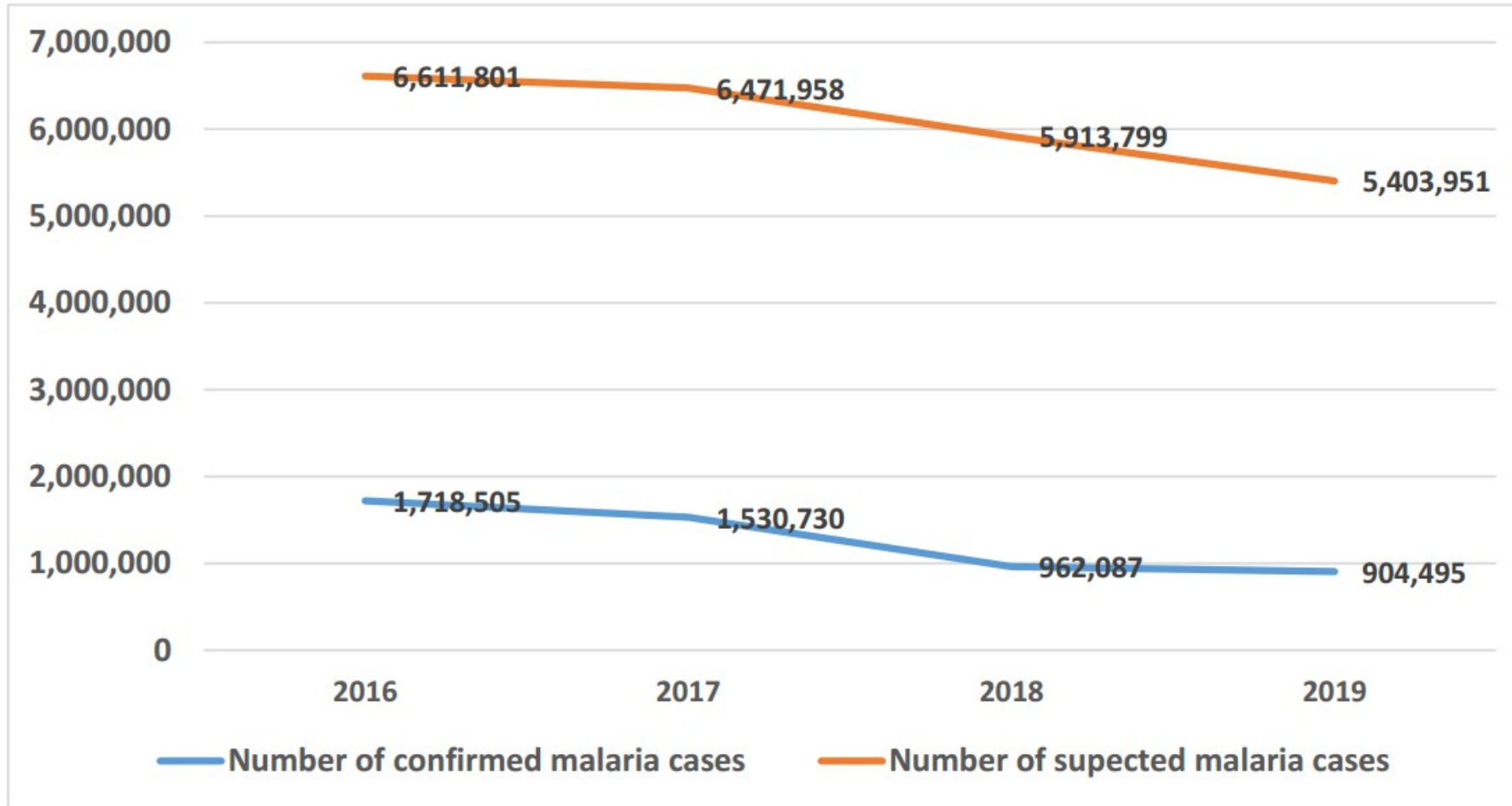


Figure 1: Trend of suspected and confirmed malaria cases, 2016-2019, NMEP, Ethiopia (taken from NMSP,2020-2025)

I. Change in estimated malaria incidence and mortality rates, 2015–2020

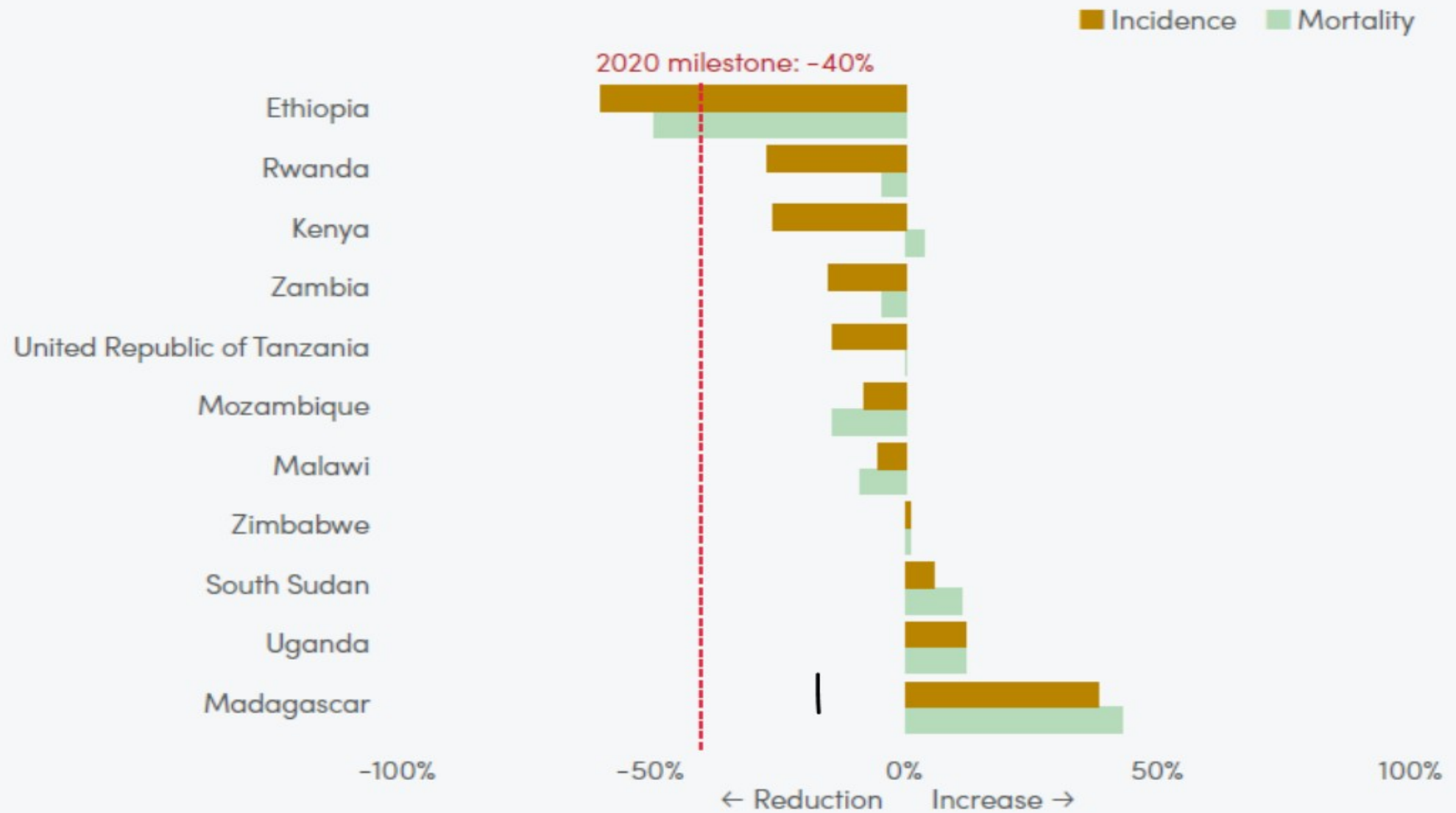
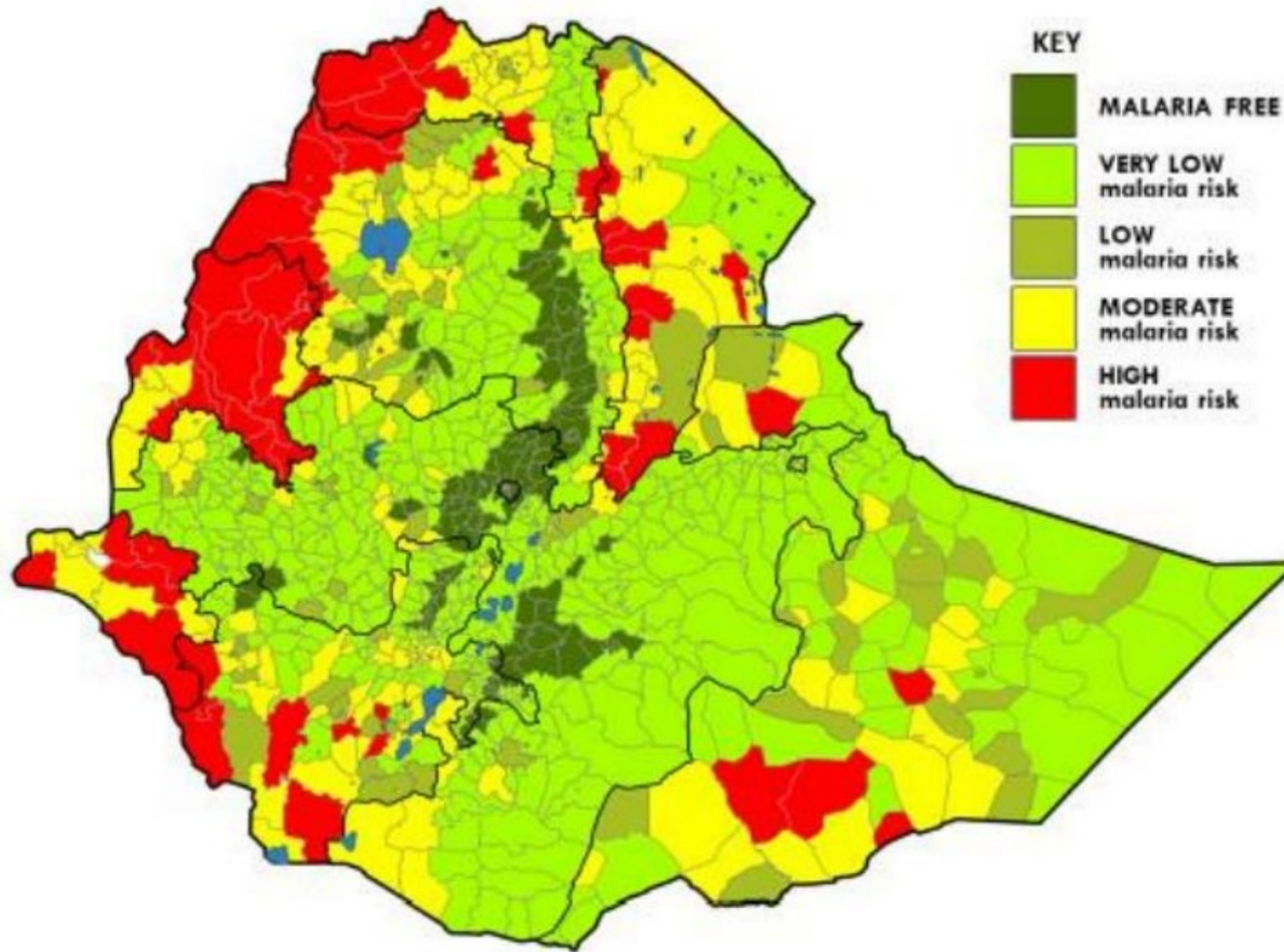


Figure 2: Reduction of malaria cases and deaths in Ethiopia (WHO 2021)



Classification of malaria transmission

- Stable (endemic):

- Transmission is high,
- could be seasonal but not subject to annual fluctuations,
- *population immunity is high.*
- Under endemic conditions, children under the age of five years, and pregnant mothers, are most likely to be infected as they have weaker immunity.

- Unstable (epidemic):

- Malaria transmission is variable, being subject to marked annual fluctuations.
- *collective population immunity is low.*
- Malaria epidemics generally occur when the population in an area *has weak immunity* to the disease because so many people in the population will be vulnerable to malaria.

Factors affecting malaria transmission

- **Altitude and climate** (rainfall, temperature and relative humidity)
- **Season**
- **Age**
- **Immunity.**

WHO, WHEN, WHERE & HOW?

❖ Everyone in malaria **belt** is at risk

- Most especially children (Under the age of 5 years)
- Pregnant women
- Non-immune adults
- Immigrants
- Travellers

Risk groups for severe falciparum malaria

- People of all ages in areas of **low endemicity** (like most endemic places in Ethiopia)
- Residents of areas where there is **little or no P. falciparum** who travel to high transmission area
- **Non-immune** pregnant women (at risk of some specific complications)
- **Internally displaced persons** moving from an area of low to an area of **high** transmission
- Children in areas of high endemicity – especially < 6 mo- 5yrs.
- People returning to highly endemic areas after a few years residence in an area with little or no falciparum malaria.
- Patients who **had a splenectomy**.

Modes of Transmission

- *Bite of female anopheles mosquito*
- Trans-placental
- Blood transfusion
- Organ transplantation

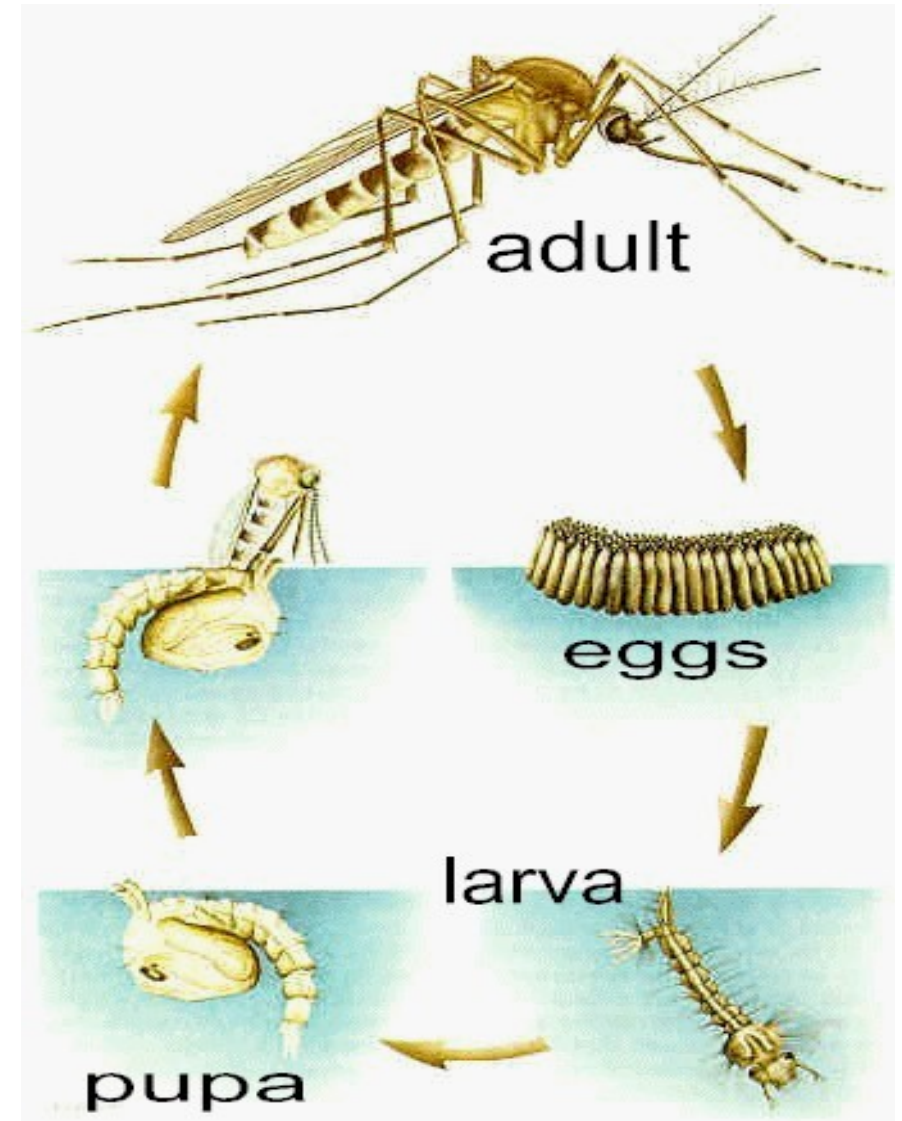


The malaria parasites and vectors

- *Plasmodium falciparum* and *P. vivax* are the two most dominant malaria parasites in Ethiopia.
- They are prevalent in all malaria-endemic areas in the country with *P. falciparum* representing about **81%** (*HMIS 2021*) of the total reported malaria cases
- *P. malariae* and *P. ovale* are rare and account for **<1%** of all confirmed malaria cases.
- The major malaria vector incriminated in Ethiopia is *Anopheles arabiensis*.

LIFE CYCLE OF MOSQUITO

- The mosquito life cycle has **four distinct developmental stages**:
 - Egg,
 - Larva,
 - Pupa and
 - Adult stages.
- **Adult mosquito** from breeding site to the house for a bite and malaria transmission



LIFE CYCLE OF MALARIA

❖ *Plasmodium* species exist in a variety of forms and have a complex life

1. human host (**asexual phase**)

A. 1st phase in hepatic cells (exo-erythrocytic phase)

B. 2nd in the red blood cells (erythrocytic phase)

2. mosquito vector (**sexual phase**)

The exoerythrocytic phase

- The **exo-erythrocytic phase** begins with inoculation of **sporozoites** into the bloodstream by a female *Anopheles* mosquito.
- Within minutes, the sporozoites enter the hepatocytes of the liver, where they develop and multiply asexually as a **schizont**.
- After **1-2 wk.** the hepatocytes rupture and release **thousands** of **merozoites** into the circulation.
- The tissue schizonts of *P. falciparum*, *P. malariae*, and apparently *P. knowlesi* rupture once and ***do not persist in the liver.***
- There are **2 types** of tissue schizonts for ***P. ovale*** and ***P. vivax***.
- The primary type ruptures in **6-9 days**, and the secondary type remains dormant in the liver cell for weeks, months, or years as **hypnozoite** before releasing merozoites and causing **relapse of infection**.

The erythrocytic phase

- The **erythrocytic phase** of Plasmodium asexual development begins when the **merozoites** from the liver penetrate **erythrocytes**.
- Once inside the **erythrocyte**, the parasite transforms into the **ring form**, which then enlarges to become a **trophozoite**.
- **The trophozoite multiplies asexually** to produce a number of **erythrocytic merozoites** that are released into the bloodstream when the erythrocyte membrane ruptures, *which is associated with fever*.

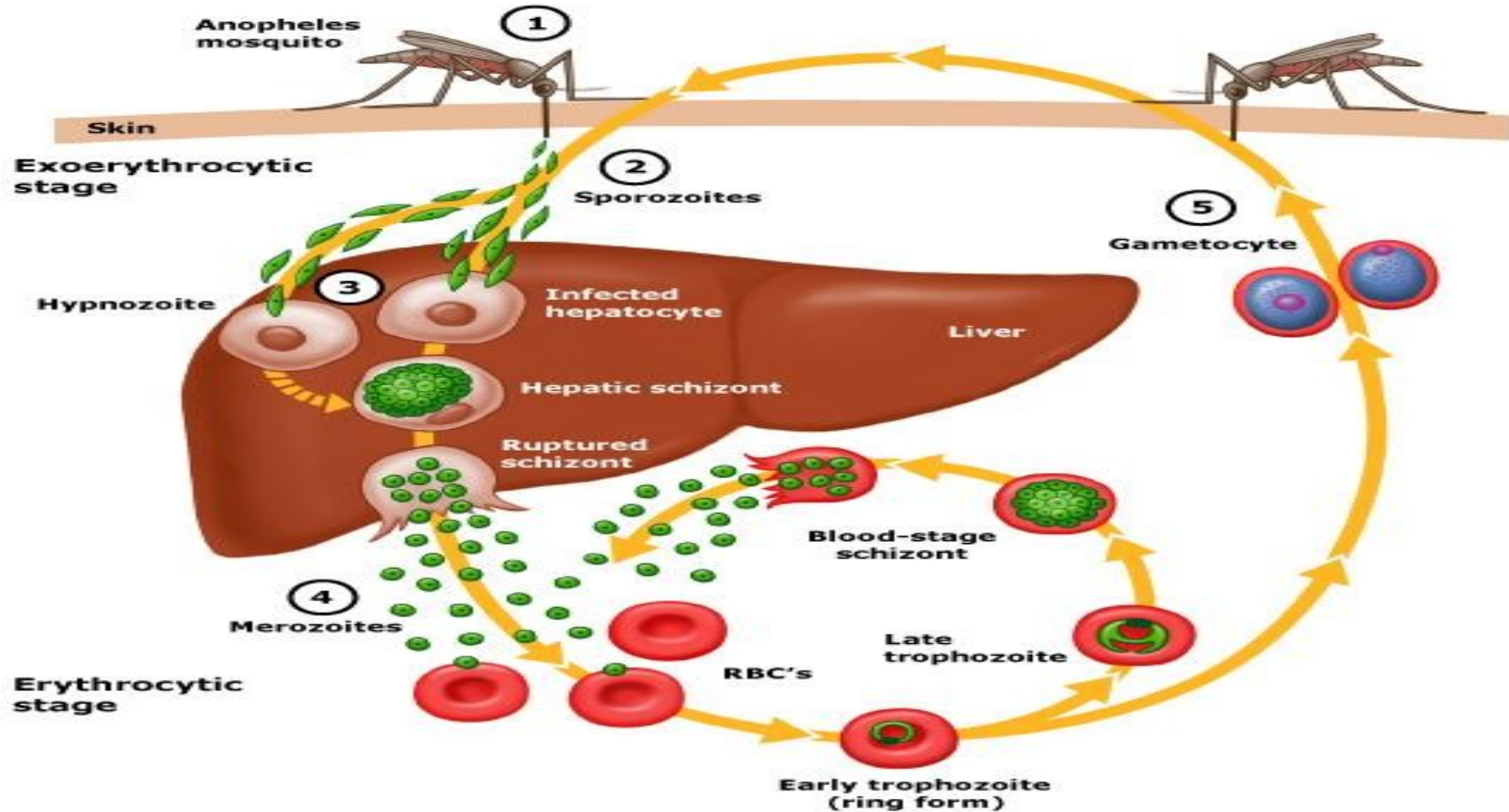
Conti

- Over time, some of the **merozoites develop into male and female gametocytes** that complete the *Plasmodium* life cycle when they are **ingested during a blood meal by the female *Anopheline* mosquito.**

Mosquito vector

- The male and female gametocytes fuse to form a **zygote** in the stomach cavity of the mosquito.
- After a series of further transformations, **sporozoites enter the salivary gland** of the mosquito and are **inoculated into a new host with the next blood meal.**

Life cycle



Video Summary ---LIFE CYCLE OF MALARIA

PATHOGENESIS

- **Four important pathogenic process**

1. **fever**

2. **anemia**

- Hemolysis, sequestration
- Dyserythropoiesis
- Shortened RBC survival
- Bone marrow suppression

Cont...

3. Immunopathologic events

- Hyper gammagloblinemia
- Immune complex formation
- Release of cytokines

4. Tissue anoxia

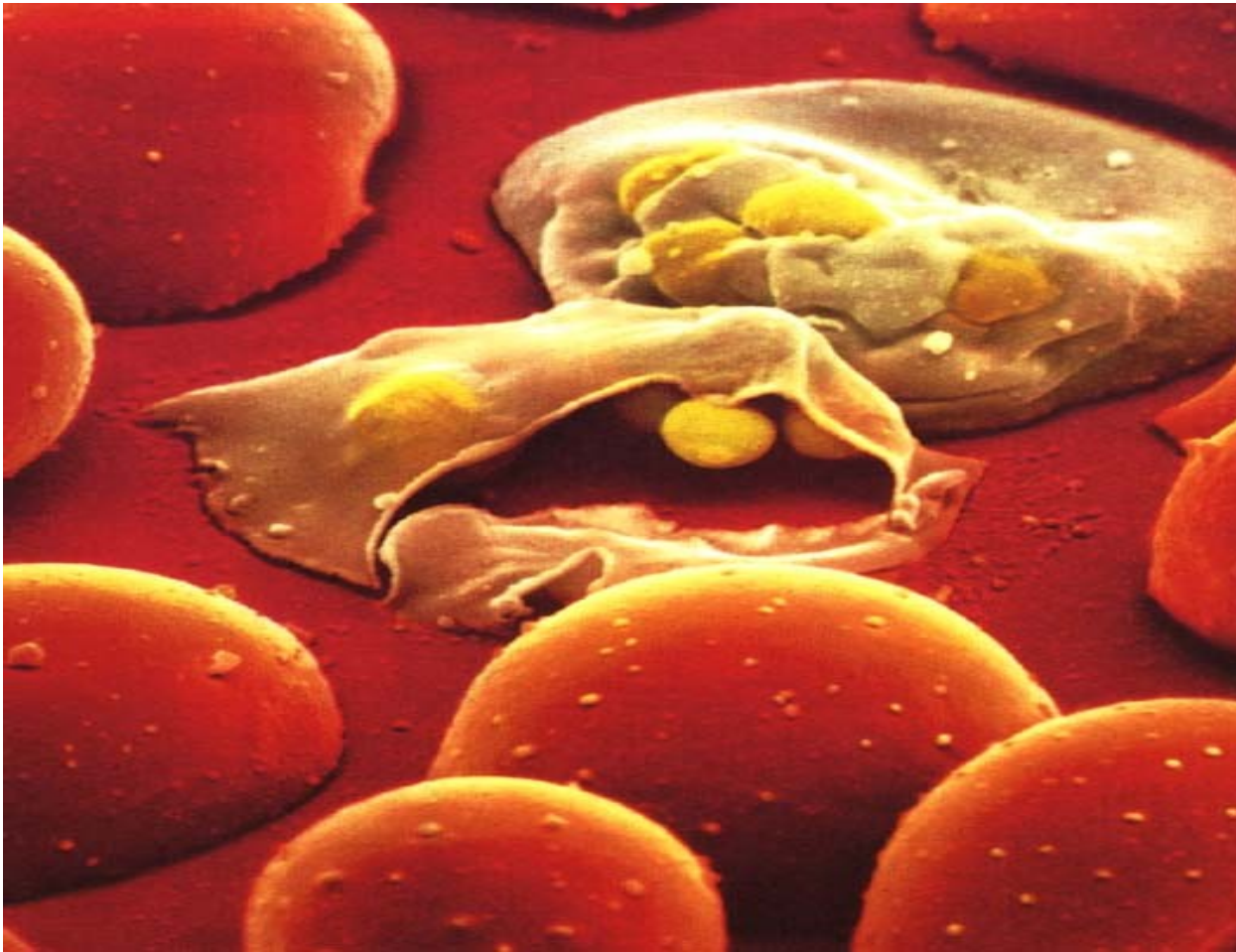
- Due to cyto adherence of infected erythrocytes
- Immunity after malarial attack is incomplete (severity decreased but prevention of further infection is unlikely)

Conti ...

- **Cytoadherence**
- **Sequestration**
- **Resetting**

Cont...

- **Severe malaria** is more common in **P. falciparum**, several processes, including
 - **higher density parasitemia**, which may lead to excessive production of proinflammatory cytokines;
 - **cytoadherence** of P. falciparum-infected erythrocytes to the vascular endothelium; and
 - **polyclonal activation**, resulting in both hypergammaglobulinemia and the formation of immune complexes.



rosette

Protection against malaria

- Acquired resistance to malaria is slow to develop and the immune and the immune mechanism involved still unclear:
- **Innate immunity:**
 - ⌚ Sickle cell trait
 - ⌚ G6PD
 - ⌚ Thalassemia trait
 - ⌚ Hereditary spherocytosis
 - ⌚ Lack of Duffy antigen (receptors of merozoites)
- **Acquired immunity:**
 - ⌚ Repeated exposure to malarial infection
 - ⌚ Neonate protected by maternal antibodies
 - ⌚ Malnutrition doesn't increase susceptibility.
 - ⌚ HIV has not been found to have a major effect on the severity of malarial disease

CLINICAL MANIFESTATIONS

- The clinical features of malaria are dependent on the malaria-specific *immune status of the host* and *the infecting species of Plasmodium*.
- Malaria is generally more severe and symptomatic in nonimmune individuals
 - From an endemic area but younger than 5 years
 - Not residing in an endemic area for more than 1 year
 - Traveler/visitor to an endemic area

Conti...

- Malaria Symptoms can develop as soon as 7 days after exposure to as late as several months after departure.
- **Classic Symptoms of Malaria are** high fever with chills, rigors, sweats, and headache, which may be paroxysmal.
- **Other manifestations as the disease progresses:** nausea, vomiting, diarrhea, cough, tachypnea, arthralgia, myalgia, abdominal & back pain.
- **Severe illness:** anemia & thrombocytopenia, along with pallor & Jaundice caused by hemolysis, Hepatosplenomegaly.
- If appropriate treatment is not administered, fever & paroxysms may occur in a cyclic pattern, depending on the infecting species.
 - Fever classically appears every other day with *P. Falciparum*, *P. Vivax*, and *P. ovale*.
 - **OR every 3rd day with *P. malariae*. [although in General practice this pattern is infrequently observed].**

Conti

- **Classical paroxysm** :

- 1. **Cold stage** – dramatic rigors/visible shakes

- 2. **Hot stage** – temperature >40 C, restless, vomiting, convulsion

- 3. **Sweating stage** – fever abates, patient may fall asleep.

- Paroxysm may last for 6-10 hours followed by a prolonged asymptomatic period, which then followed by further rigors in untreated patient.

Cont...

- ***P. falciparum*** is the most severe form of malaria and is associated with more intense parasitemia, **why?**
 - Falciparum infects all ages of red cell
 - Falciparum can lead to far greater parasitemia
 - Falciparum sequestrates-vivax, malariae and ovale do not
- Fatality rates of up to **25%** in non-immune adults and **30%** in non-immune infants occur if appropriate therapy is not instituted promptly

Congenital malaria

- The 1st sign/symptom most commonly occurs between 10 and 30 days of age (range 14 hr to several months of age)
- Usually associated with placental parasitemia
- Fever ,irritability , refusal of feeds
- Anemia, jaundice and hepatosplenomegaly
- Mother may have placental parasitemia, peripheral parasitemia or both with out any fever or other clinical manifestations
- Vertical transmission is as high as 40%

Uncomplicated malaria

- The initial symptoms of malaria are nonspecific and may also include
 - Tachycardia & tachypnea,
 - chills, malaise, fatigue & diaphoresis,
 - Headache & cough,
 - anorexia, nausea & vomiting,
 - abdominal pain & diarrhea,
 - Arthralgias & myalgias
- In general, patients who are not obtunded and can take medicines orally have "uncomplicated" disease.

Complicated malaria

❖ Severe malaria

- **Definition:** The presence of one or more signs and symptoms of severe illness and demonstrable asexual form of Falciparum.
- All age is at risk but More common in under five child and pregnant women.
- Although most severe and complicated malaria is usually due to P. falciparum, patients with complicated infections due to P. vivax have also been described.

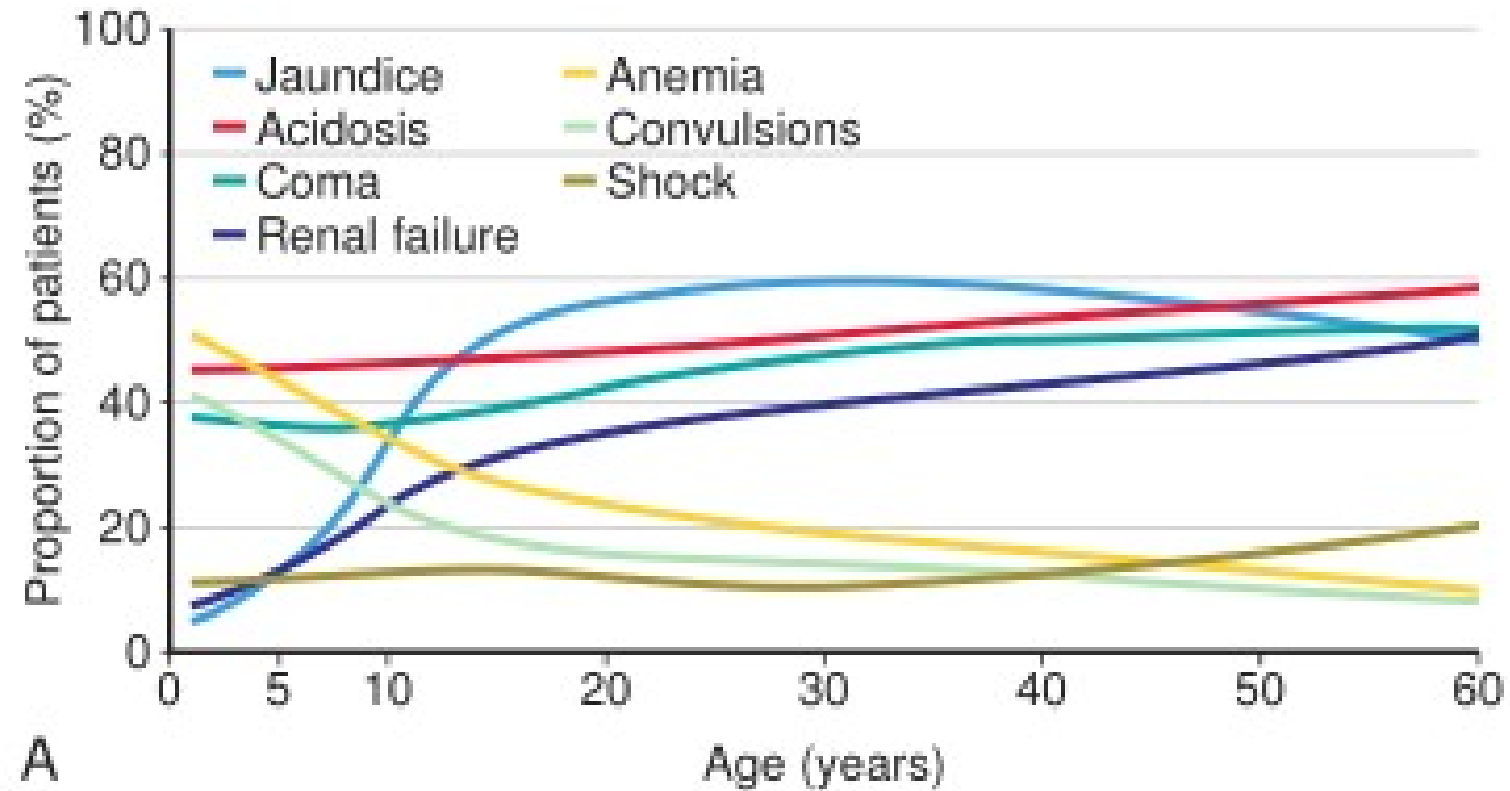
Complicated Severe malaria

1. Impaired consciousness
2. Prostration
3. Respiratory distress
4. Multiple seizures
5. Jaundice
6. Hemoglobinuria
7. Abnormal bleeding
8. Severe anemia
9. Circulatory collapse(algid malaria)
10. Pulmonary edema

Table 6 Severe manifestation of malaria and frequency in adults and children

	Frequency in	
Clinical features¹	Children	Adults
Prostration (unable to sit unsupported (>1yr) or inability to drink or breastfeed (<1yr)	+++	+++
Impaired consciousness or un-rousable coma	+++	++
Respiratory distress (acidotic breathing/deep breathing or in-drawing of chest wall)	+++	+
Multiple convulsions	+++	+
Circulatory collapse or shock	+	+
Pulmonary edema or difficulty in breathing	+/-	+
Bleeding tendency/abnormal bleeding	+/-	+
Jaundice	+	+++
Haemoglobinuria	+/-	+
Laboratory findings		
Severe anemia (hemoglobin <5g/dl, haemocrit < 15%)	+++	+
Hypoglycemia (<2.2 mmol/L or 40 mg/dL)	+++	++
Acidosis (bicarbonate <15 mmol/L)	+++	++
Hyperlactatemia (>5 mmol/L)	+++	++
Hyperparasitemia (>2%) (non-immune person)	++	+
Renal impairment (creatinine >265 umol/L)	+	+++

Prevalence of severe malaria manifestations based on age



A

DIAGNOSIS

1. Clinical diagnosis---- high index of suspicions

- ❖ By taking an *accurate history* of the illness and *physical examination*.
- ❖ **Clinical diagnosis** of malaria is made in a patient *who has fever* or *history of fever* in the **last 48 hours** and lives in malaria-endemic areas or has a history of travel within the last 30 days to malaria-endemic area

Cont...

2. Parasitological test

- Microscopy (blood film)
- RDT(rapid diagnostic tests)

Microscopy (blood film).....Gold standard

A. Thick film

- The concentration of erythrocytes on a thick smear is 20–40 times that on a thin smear
- used to quickly scan large numbers of erythrocytes

B. Thin smear

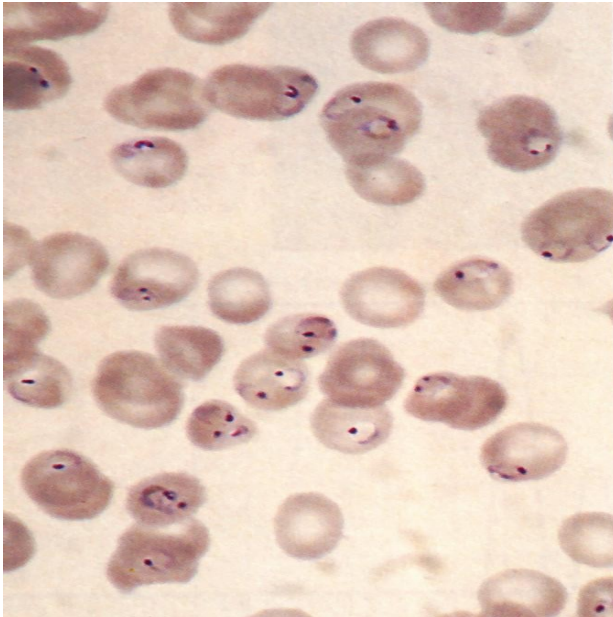
- for identification of the malaria species
- determination of the percentage of infected erythrocytes
- useful in following the response to therapy
- A single negative blood smear does not exclude malaria

Microscopy

Thin film

Fix, stain

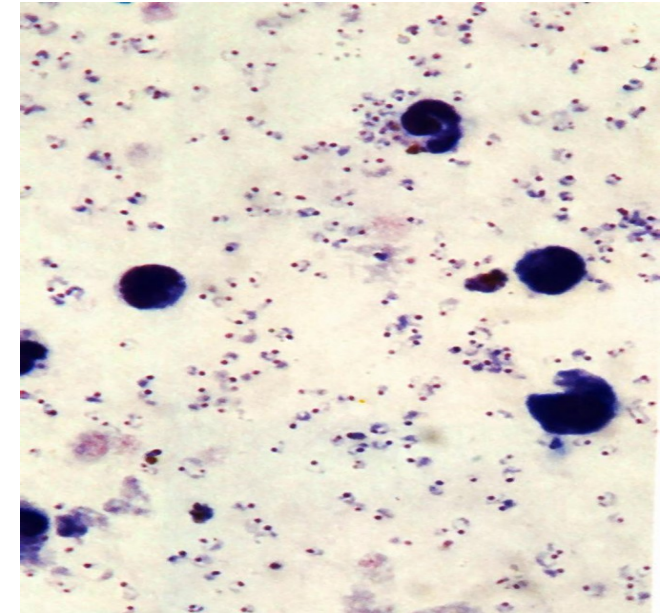
Species identification



■ Thick film

□ Stain, haemolyse

□ concentration



Which smear is the most sensitive measure to detect low-density infection?

The thick smear is the **most sensitive** measure to detect low-density infection because it allows the microscopist to review a large volume of blood and is read for detection of infection.

Cont....

Rapid diagnostic tests(RDTs)

- Detect antigen or enzyme...
- Positivity after cure (2 wks)
- Can't replace microscopy but can be supplementary
- High sensitivity (>95%)
- Stable in all situations
- Not of much usefulness in **high transmission areas**
- based on the detection of parasite-derived histidine-rich protein 11 (HRP11), parasite lactic dehydrogenase(LDH), or aldolase

Conti

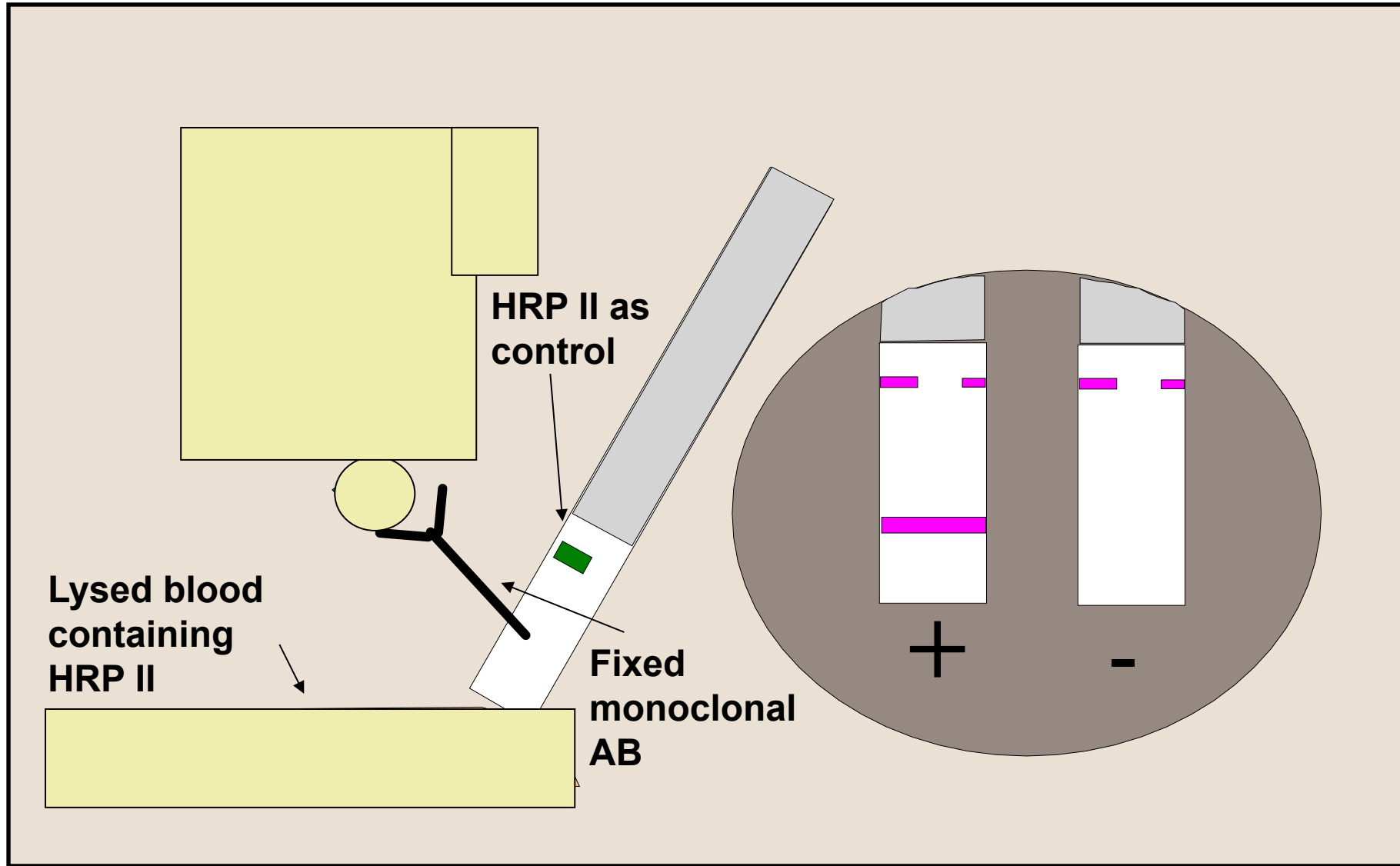
- Currently **multi-species RDTs** capable of specifically detecting both *P. falciparum* and *P. vivax*, are being supplied by FMOH to health posts.
- Sensitivity and specificity for *P. falciparum* (94-99% and 94-99%, respectively) and *P. vivax* (87-93% and 99%, respectively) are good, but sensitivity for *P. ovale* and *P. malariae* is lower.
- Patients who test negative by malaria RDT or microscopy **do not need** anti-malarial medications.

Conti...

- Inability, in the case of PfHRP2-based RDTs, to distinguish new infections from recently and effectively treated infections, due to the persistence of the antigen in the blood 1-5 weeks after effective treatment.
- The emergence of Falciparum strains with HRP2 deletions.
- The heterogeneous quality of available RDT kits

Rapid Diagnostic Tests (RDT)

Current tests all immuno-cromatographic tests



RDT interpretation chart

Results	Control Line	Test Lines	
		<i>P.falciparum</i>	<i>P.vivax</i>
Negative			
Positive <i>P.falciparum</i> only			
Positive <i>P.falciparum</i> only combined with other species			
Positive non- <i>P.falciparum</i> (<i>P.vivax</i>)			
Invalid			
Invalid			
Invalid			

Laboratory tests

- Blood glucose level
- CBC
 - Hemoglobin (Hb) estimation or packed cell volume (hematocrit)
 - Leukocyte count
 - Platelet count
- Lumbar puncture (if convulsions or change in mentation)
- BUN & creatinine
- Liver function test

Investigation

Investigation	
Blood film for Malaria Parasite (BFMP)	-to be done immediately, if initial film is negative, repeat 2 or more
Blood Urea & Serum Electrolyte (BUSE)	-hyponatremia -hypocalcemia -hypoalbuminemia
C-reactive protein (CRP)	-raised
Liver function test (LFT)	-hyperbilirubinemia -mild elevation of liver enzyme
Full blood count (FBC)	-WBC generally normal -lymphocytopenia -thrombocytopenia
Blood glucose	Can be low in severe cases

Box 1

Differential diagnoses of malaria

Sepsis due to bacteremia
Encephalitis (rickettsial or viral)
Meningitis (bacterial or viral)
Pneumonia (bacterial, viral or fungal)
Typhoid fever
Dengue fever
Chikungunya
Leptospirosis
Brucellosis
Rickettsial infections
Acute schistosomiasis (Katayama fever)
Amebic liver abscess
Acute HIV

Malaria differential diagnosis

Non complicated

- Relapsing fever
- Typhus fever
- Typhoid fever
- Fever with focal sign (urinary tract infection, respiratory tract infections, otitis media, sinusitis, tonsillitis, osteomyelitis, arthritis, dysentery)
- Influenza
- Visceral leishmaniasis

Complicated Malaria

Decreased Level of Consciousness

- Viral encephalitis
- Bacterial meningoencephalitis
- Cerebral typhoid
- Cerebro-vascular event
- Complicated typhus, relapsing fever
- Febrile illness with hypoglycaemia
- Sepsis
- Convulsion in a patient with fever

Renal failure

- Glomerulonephritis
- Acute tubular necrosis due to hypovolemia or hypotension

Uncomplicated malaria Treatment

- **P. falciparum** positive by multi-species RDT: **AL and single dose primaquine** (sdPQ) is the recommended first-line drug.
- AL tablets are given according to body weight in six doses over three days.
- Treat uncomplicated P. falciparum malaria in pregnant women including *first trimester* and *breastfeeding* women with **AL**.
- **Treat infants weighing < 5 kg** with uncomplicated P. falciparum malaria with AL at the same mg/kg body weight target dose as for children weighing 5 kg.
- The first dose should be given under the direct supervision of the health worker.
- **AL** should preferably be taken with food or fluids. A **fatty meal** or milk improves the absorption of the drug.

Conti...

- AL is available in co-formulated tablets containing **artemether 20 mg** and **lumefantrine 120 mg** per tablet.
- To reduce the transmission of *P. falciparum* infection give a single dose of **0.25 mg/kg** body weight primaquine with AL (except *pregnant women, infants aged < 6 months* and *women breastfeeding infants aged < 6 months*).
- Testing for G6PD deficiency is not required for single dose primaquine treatment.

Second-line treatment

- **Second-line treatment** is a drug used after failure of first-line treatment or in patients who are allergic to or unable to tolerate the first-line treatment.
- In Ethiopia **dihydroartemisinin-piperaquine (DP)** is recommended as a second-line treatment for both **P. falciparum** and **P.vivax**.
- It has a longer prophylactic effect for P.vivax compared to AL.
- Dihydroartemisinin-piperaquine (DP) is a new co-formulated ACT and is well tolerated.
- It is proved to be highly effective against uncomplicated **P. falciparum** and **P.vivax** malaria.
- The drug is WHO approved and registered in Ethiopia. It is available in global market.
- It has pediatric and adult preparations.

CONTI,,,

- P. vivax, and malaria species other than P. falciparum positive by RDT: **The first-line drug of choice is chloroquine plus 0.25mg/kg between primaquine radical cure for 14 days.**
- *Though the prevalence of G6PD deficiency is very low in Ethiopia*, health workers should closely follow-up patients started on primaquine for hemolysis by informing the patient the symptoms of hemolytic anemia
- If there is evidence of hemolysis, primaquine should be discontinued and should not be given to the patient in the future.
- Chloroquine preparation is 150 mg base tablet OR chloroquine syrup 50 mg base (Annex D for recommended dosage).
- A 250 mg tablet of chloroquine phosphate (“salt”) is the same as chloroquine 150 mg base.
- If chloroquine syrup is not available, the alternative treatment is AL for under-five children.

Conti,,,

- **Multi-species RDT positive for *P. falciparum* and *P. vivax* (mixed infection):**
- The recommended first-line treatment for mixed infection is **AL and radical cure PQ.**
- Primaquine is contraindicated in pregnancy. To prevent **relapses** of *P. vivax* in pregnant women, CQ treatment should be followed by weekly CQ prophylaxis until delivery and six months of breastfeeding. Then PQ radical cure should be given if there is no contraindication.
- **Multi-species RDT negative for malaria:** If the result of the multi-species RDT is negative for all malaria species, **malaria is unlikely**. Refer the patient to the next higher-level health facility.

Alternate treatment for uncomplicated malaria:

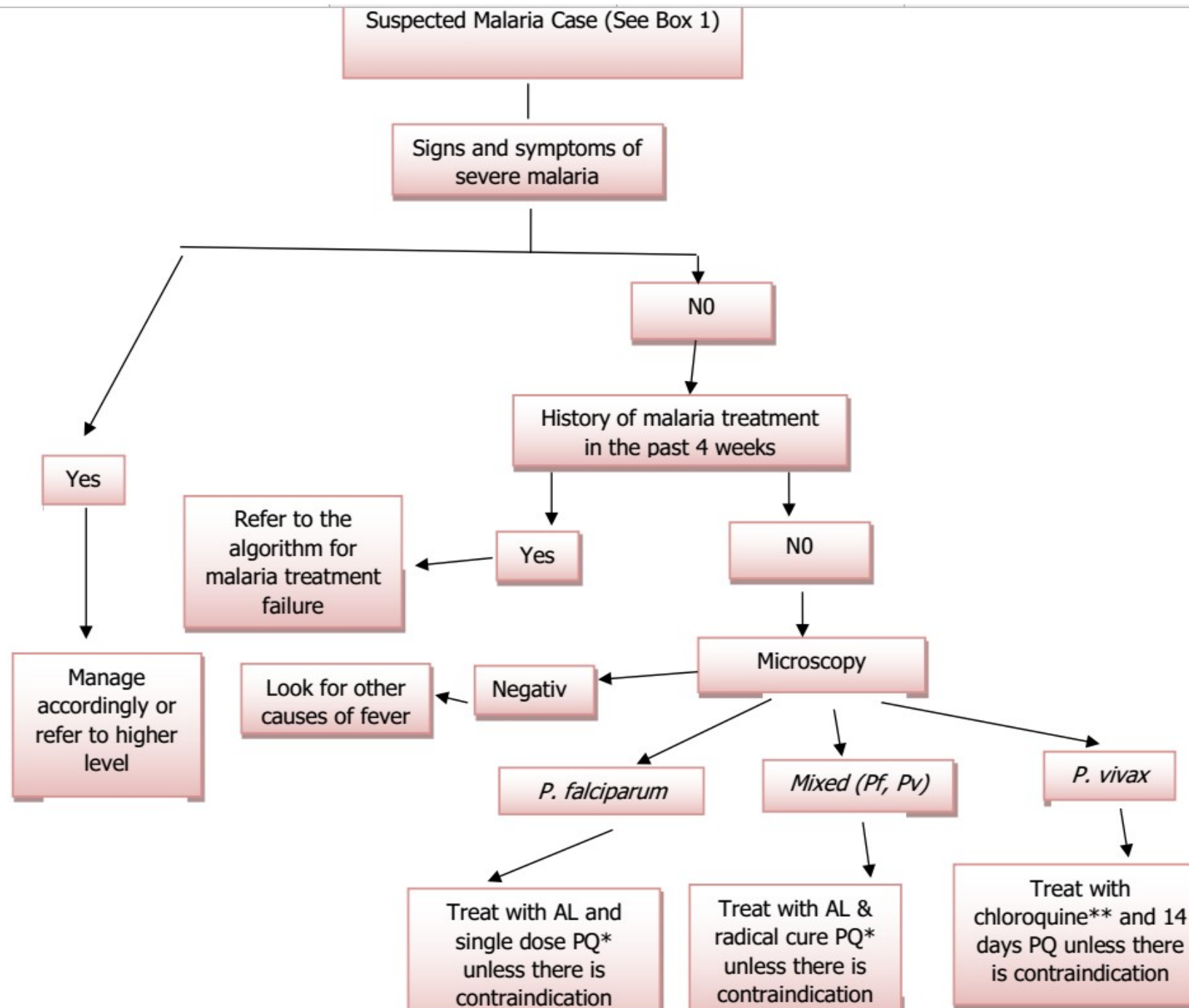
- **AL** may be used to treat *P. vivax* infection when chloroquine is not available.
- If AL is not available for *P. falciparum* or mixed malaria infections, use the second-line drug which is *dihydroartemisinin piperaquine* (**DP**).
- If both **chloroquine** and **AL** are not available for *P. vivax* infection, use the second-line drug (**dihydroartemisinin piperaquine**).
- If quinine tablet is the only antimalaria drug available, it can be used to treat uncomplicated *P. vivax* or *P. falciparum* infections.

Table 8. Recommended first-line antimalarial drugs at Health Post and Health Center/Hospital

Parasite	Health post (RDT)	Health center/Hospital (microscopy)
P. vivax, without contraindication to PQ*	CQ + PQ (radical cure)	CQ + PQ (radical cure)
P. vivax in pregnant women	CQ treatment should be followed by weekly CQ prophylaxis until delivery and six months of breastfeeding. Then give PQ radical cure if there is no contraindication	CQ treatment should be followed by weekly CQ prophylaxis until delivery and six months of breastfeeding. Then give PQ radical cure if there is no contraindication
P. vivax with contraindication to PQ other than pregnancy	CQ	CQ
Uncomplicated P. falciparum malaria without contraindication to PQ	AL + PQ (single dose)	AL + PQ (single dose)
Uncomplicated P. falciparum in pregnancy including first trimester**	AL	AL

Uncomplicated <i>P. falciparum</i> in pregnancy including first trimester**	AL	AL
Uncomplicated <i>P. falciparum</i> with contraindication to PQ other than pregnancy	AL	AL
Uncomplicated mixed infection without contraindication to PQ*	AL + PQ (radical cure)	AL + PQ (radical cure)
Uncomplicated mixed infection in pregnancy including first trimester**	AL treatment should be followed by weekly CQ prophylaxis until delivery and six months of breastfeeding. Then give PQ radical cure if there is no contraindication	AL treatment should be followed by weekly CQ prophylaxis until delivery and six months of breastfeeding. Then give PQ radical cure if there is no contraindication
Uncomplicated mixed infection with contraindications to PQ other than pregnancy	AL	AL

*Contraindications to radical cure primaquine: Pregnancy, in breastfeeding mothers less than six months infants, infants under six months, moderate to severe anemia, known hypersensitivity to primaquine and known history of G6PD deficiency.



Advantages of ACT

- **Why ACT?:**
 - Rapid clinical response...
 - Effect on parasite biomass...
 - Reduction of gametocytaemia
 - Efficacy ..
 - Few reported adverse effects...

Annex C. Tablet containing 20 mg Artemether plus 120 mg Lumefantrine in a fixed dose.

Weight (KG)	Day 1		Day 2		Day 3		Total tabs	Color code
	Immediate	After 8 hours	Morning	Evening	Morning	Evening		
<14kg	1 tablet	1 tablet	1 tablet	1 tablet	1 tablet	1 tablet	6	Yellow*
15-24 kg	2 tablets	2 tablets	2 tablets	2 tablets	2 tablets	2 tablets	12	Blue*
25-34 kg	3 tablets	3 tablets	3 tablets	3 tablets	3 tablets	3 tablets	18	Brown
>35	4 tablets	4 tablets	4 tablets	4 tablets	4 tablets	4 tablets	24	Green

* (Yellow, Blue) flavored pediatric formulation (dispersible tablets) of Artemether-Lumefantrine (AL) is available for enhancing its use in young children.

Contraindications:

- ❖ Persons with a previous history of reaction after using the drug;
- ❖ Pregnant women in the first trimester

Persons with severe and complicated malaria should not be treated with oral medications.

Treatment of severe malaria

- Admit the patient
- Maintain airway, breathing and circulation
- Insert NG tube for feeding and medication
- Urinary catheter/glove catheter for UOP measurement
- Establish an IV infusion
- Take blood sample while establishing an IV line for
 - Glucose
 - Blood film
 - hematocrit

Cont'd...

- First line treatment for severe malaria due to *P. falciparum* is either:
 - IV or IM artesunate (preferred) *OR*
 - IM artemether (alternative) *OR*
 - IV quinine infusion (if artesunate is not available) *OR*
 - IM quinine
- Artesunate 2.4 mg/kg IV or IM given on admission (time = 0), then at 12 hrs and 24 hrs, then once a day for 5-7 days is the recommended treatment.

- Severe Malaria

- **artesunate** 2.4 mg/kg body weight IV or IM given on admission (time = 0), then at 12 h and 24 h, then once a day
- **quinine** 20 mg salt/kg body weight on admission (IV infusion or divided IM injection), then 10 mg/kg body weight every 8 hr.
- **artemether** 3.2 mg/kg body weight IM given on admission then 1.6 mg/kg body weight per day should only be used if none of the alternatives are available as its absorption may be erratic.

Conti,,,

- Artesunate is the preferred first-line parenteral treatment for severe malaria in Ethiopia.
- IV or IM artesunate has been shown to reduce the risk of death from severe malaria significantly by **24%** in children and **39%** in adults (Artesunate versus quinine for treating severe malaria:

Cont'd...

- If artesunate or artemether is not available give IV quinine
- For all patient with severe malaria quinine should be given at least for the first 24 hr Loading dose 20mg/kg
- Avoid loading dose if the patient received preceding quinine or mefloquine within the preceding 7 days.
- If required for more than 48 hr decrease the dose $\frac{1}{3}^{\text{rd}}$ - $\frac{1}{2}$
- If required more than 4-5 days infusion rate should not exceed 5mg/kg/hr.

Cont'd...

- Frequent RBS , Hct, renal function test & electrolyte should be checked otherwise glucose should be given
- Frequent monitoring of therapeutic response with both parasitologic and clinical by monitoring vital sign, urine output, mental status etc.
- If temp >39 give antipyretics like paracetamol.

COMPLICATIONS

Cerebral malaria

- Defined as the presence of coma in a child with *P. falciparum* parasitemia and an absence of other reasons for coma.
- The onset may be gradual or sudden following a convulsion.
- Focal neurologic signs are unusual.
- Cerebral malaria is associated with a fatality rate of **20-40%** and is associated with long-term cognitive impairment in children.

Cont'd...

- Hypoglycemia is common, but children with true cerebral malaria fail to arouse from coma even after receiving a dextrose infusion.
- Among survivors of cerebral malaria, neurologic sequelae are more common among children than adults (about 15 versus 3 percent, respectively).
- Residual deficits may include hemiplegia, cerebral palsy, cortical blindness, deafness, epilepsy, language deficit, and impaired cognition.

Cont'd...

- **Risk factors** for cerebral malaria include:
 - Age (children and elderly),
 - Pregnancy,
 - Poor nutritional status,
 - HIV infection,
 - Host genetic susceptibility, and
 - History of splenectomy
 - First attack

Cont'd...

- Treatment is largely supportive and includes evaluation and treatment of seizures and hypoglycemia.
- Although increased intracranial pressure has been documented in some children with cerebral malaria, treatment with mannitol and corticosteroids has not improved outcomes in these children.

Convulsions

- Ensure the patient is in a safe environment.
- Do not pin the patient down or try to put something in his mouth.
- Try to get the patient onto his side to protect the airway or gently thrust the jaw forward.
- Diazepam 0.15 mg/kg body weight IV OR 0.5 mg/kg body weight rectally

Anemia

- Anemia associated with malaria is partly due to the destruction of red cells that contain parasites (hemolysis).
- Several other mechanisms may accelerate the development of anemia:
 - Non-parasitized red cells are destroyed more quickly than normal cells.
 - Increased splenic sequestration
 - Bone marrow does not function adequately.
 - Anemia is worsened if there is abnormal bleeding, intravascular hemolysis or renal failure.
 - Repeated infection

Cont'd...

- The most common clinical sign of anemia is pallor of the palms and is the most reliable sign for detecting anemia in children.
- Indication for blood transfusion
 - Hemoatcrit below 15%
 - Anemia associated with acidosis, shock or high parasitemia
- Rx = 10 ml of packed cells OR 20 ml whole blood/kg of body weight.

Acute renal failure

- Acute renal failure is due to acute tubular necrosis & it is a common complication in adults, but is rarely seen in children.
- It is worsened by hypovolemia and hypotension.
- It is highly preventable if fluid balance and blood pressure is maintained.
- Management: Correct dehydration and maintain fluid balance, monitor JVP, do peritoneal dialysis.

HYPOGLYCAEMIA

- Patient with severe disease especially young children
- Common in patient in coma & convulsion
- Quinine induced hyperinsulinaemia
- Anxiety sweating, dilatation of pupil ,breathlessness, laboured and noisy breathing ,feeling of cold tachycardia and light headedness
- It is easily overlooked clinically because the manifestations may be similar to those of cerebral malaria.
- 10%glucose to prevent recurrence of hypoglycemia

Pulmonary edema and ARDS

- ARDS appears to be due to the direct effect of parasites sequestered in the lungs, possibly through release of cytokines.
- It is indistinguishable from pulmonary edema but both of these complications are unusual in children.
- Hyperventilation (rapid breathing) is the initial manifestation.
- Crackles are present on auscultation, and pink frothy sputum (severe cases).

Cont'd...

■ Management

- Position patient upright (sitting position)
- Give oxygen therapy
- Give diuretics e.g. furosemide 1 mg/kg IV.
- If no response increase dose progressively to maximum 6mg/kg/day
- Assess the need for intubation and mechanical ventilation including positive end expiratory pressure (PEEP)
- Perform regular suction

Spontaneous bleeding (DIC)

- In *P. falciparum* malaria, the platelet count is typically reduced.
- Nevertheless, spontaneous bleeding is rare in both children and adults.
- When it develops, it results from disseminated intravascular coagulation (DIC).
- RX transfuse fresh blood ,clotting factors or platelets as required

Shock

- Shock is due to inadequate cardiac output and poor tissue perfusion.
- In some patients it may occur concurrently with bacteremia.
- Shock is not usually associated with malaria alone, and therefore, additional bacteremia should be suspected.
- Low blood pressure; feeble pulse; impaired tissue perfusion with cold clammy skin and peripheral cyanosis.

Cont'd...

- In children delayed capillary refill is a useful sign; signs of dehydration.
- Management:
 - Maintain fluid balance, administer 20 ml/kg fluid bolus
 - Check for bacteremia (blood cultures, WBC)
 - If indicated give appropriate antibiotic
 - Monitor vital signs.

Hyperparasitemia

- Parasite density $>5\%$ or ++++ and peripheral schizontaemia
- Easily leads to anemia in children
- Treatment
 - Initial loading dose of parental anti malarial drug even if patient tolerate medication by mouth

Hemoglobinuria

- Hemoglobinuria results from the rapid breakdown of red blood cells (massive intravascular hemolysis) in the circulation.
- The urine is dark, and tests strongly positive for blood (Hb) but contains no red cells on microscopy.
- The plasma may also be dark because of the hemoglobin released from the red cells.

Cont'd...

- **Management:**

- Maintain hematocrit above 15%
- Monitor JVP to avoid fluid overload and hypovolaemia
- If oliguria develops and blood urea and serum creatinine levels rise, consider peritoneal dialysis or hemodialysis
- Continue anti-malarial therapy.

Chronic complications of malaria

1. Hyper reactive malarial splenomegaly(HMS)

- HMS is most common in young and middle-aged adults
- HMS is rare in children younger than 8 years but was reported in a 3-year-old patient.
- The most common presenting symptoms are chronic abdominal swelling (64%) and pain (52%).
- Weight loss
- Intermittent fever is rare

Diagnostic criteria

- **Major criteria**

- Gross splenomegaly 10 cm or more below the costal margin in adults for which no other cause can be found
- Elevated serum IgM level 2 standard deviations or more above the local mean
- Clinical and immunologic responses to antimalarial therapy
- Regression of splenomegaly by **40%** by 6 months after start of therapy
- High antibody levels of *Plasmodium* species ($\geq 1:800$)

- **Minor criteria**

- Hepatic sinusoidal lymphocytosis
- Normal cellular and humoral responses to antigenic challenge, including a normal phytohemagglutination response
- Hypersplenism
- Lymphocytic proliferation
- Familial occurrence

Cont'd...

- Treatment

- Chloroquine 300mg po weekly for 3-6 months. Wkly dosing and daily dosing are said to be equally effective.
- monitor the patient for ophthalmologic effects with slit lamp, funduscopic, and visual field examinations.

- Splenectomy plays no role

Cont'd...

2. Quartan Malarial nephropathy

- Repeated *P. malariae* infection → soluble immune complex deposition in glomerular BM → Nephrotic syndrome (FSGS)
- Treatment: Poorly responsive to antimalarial or steroid + cytotoxic drugs

3. Burkitt's lymphoma + EBV infection

- High prevalence in malarious areas of Africa
- malaria induced immuno suppression may provoke EBV infection

Poor prognosis

■ Clinical indicators

- Age under 3 years
- Deep coma
- Witnessed or reported convulsions
- Absent corneal reflexes
- Decerebrate/decorticate rigidity or opisthotonos
- Clinical signs of organ dysfunction (e.g. renal failure, pulmonary oedema)
- Respiratory distress (acidosis)
- Circulatory collapse
- Papilloedema and/or retinal oedema

Cont'd...

■ **Laboratory indicators**

- Hyperparasitaemia ($>250\,000/\mu\text{l}$ or $>5\%$) (see page 25)
- Peripheral schizontaemia
- Peripheral blood polymorphonuclear leukocytosis ($>12\,000/\mu\text{l}$)
- Mature pigmented parasites ($>20\%$ of parasites)
- Packed cell volume less than 15%
- Haemoglobin concentration less than 5 g/dl
- Blood glucose less than 2.2 mmol/l (<40 mg/dl)

Cont'd...

- Blood urea more than 60 mg/dl
- Serum creatinine more than 265 $\mu\text{mol/l}$ (>3.0 mg/dl)
- High CSF lactic acid (>6 mmol/l) and low CSF glucose
- Raised venous lactic acid (>5 mmol/l)
- More than 3-fold elevation of serum enzymes (aminotransferases)
- Increased plasma 5'-nucleotidase
- Low antithrombin III levels
- Very high plasma concentrations of tumour necrosis factor (TNF)

PREVENTION& CONTROL

➤ Strategic approaches:

- Early diagnosis and effective treatment
- Vector control :
 - Insecticide treated materials(ITN)
 - Residual house spray
 - Other vector control methods; Environmental Mx, Larviciding etc
- Epidemic prevention and control
- Chemoprophylaxis

PREVENTION

How can malaria be prevented?

1. **Vector control interventions.** Vector control is the main approach to prevent malaria and reduce transmission.

Two forms of vector control are effective for people living in malaria-endemic countries:

- ✓ **insecticide-treated nets**
- ✓ **indoor residual spraying**

For travellers, the use of an insecticide-treated net is the most practical vector control intervention.

WHO maintains a list vector control products that have been assessed for their safety, effectiveness and quality.



Conti,,,

2. Chemo preventive therapies and chemoprophylaxis.

Although designed to treat patients already infected with malaria, some antimalarial medicines can also be used to prevent the disease.

- ✓ Current WHO-recommended malaria chemopreventive therapies for people living in endemic areas include
- ✓ intermittent preventive treatment of malaria in pregnancy
- ✓ perennial malaria chemoprevention
- ✓ seasonal malaria chemoprevention
- ✓ post-discharge malaria chemoprevention, and intermittent preventive treatment of malaria for school-aged children.

Chemoprophylaxis drugs are also given to travellers before entering an area where malaria is endemic and can be highly effective when combined with insecticide-treated nets.

Prophylaxis

- Chemoprophylaxis is necessary for all visitors to and residents of the tropics who have not lived there since infancy
- Children of nonimmune women should have chemoprophylaxis from birth.
- Chemoprophylaxis should be started 1–2 wk before entering the endemic area (except for doxycycline administration, which can begin 1–2 days before departure) and continue for at least 4 wk after leaving

Conti ...

Give drugs based on sensitivity patterns

- Chloroquine
- Mefloquine
- Doxycycline
- Atovaquone/proguanil

Reference

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2. UpToDate 2023
3. National Malaria Guidelines 5th edition 2022.
4. WHO Guidelines for malaria 2023
5. Red Book:Report of the Committee on Infectious Diseases 32nd edition
6. Malaria Consortium - Pneumonia Control Strategies in Chad and Ethiopia saving lives; <https://www.malariaconsortium.org/pages/pneumonia-control-strategies-in-chad-and-ethiopia-saving-lives.htm>

Reading Assignment

- Typhoid Fever
- Relapsing Fever
- Endemic Typhus
- Schistosomiasis
- Leishmaniasis

END

THANK YOU !!!